


# Anterior Ceramic Crowns

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The provision of anterior ceramic crowns can be the most valuable and difficult service in dentistry. Protecting the natural dentition and providing an illusion of reality requires the practitioner to choose the correct ceramic system and prepare the site for success. This chapter discusses a systematic approach for selection of the proper ceramic system[1] and development of a foundation for the restoration that enables predictable results.

## Decision-Making Factors

There are two common myths pertaining to anterior ceramics: *(1)* Strength is the most important decision-making factor, and *(2)* all-ceramic crowns are always more esthetic.

Strength should not be the overriding factor in choosing an anterior ceramic crown system. While it is true that metal-ceramic crowns are stronger than layered all-ceramic systems, the real question is how much strength is required. The answer is that the crown must be able to resist fracture under load. Normal incisal bite force averages between 100 and 300 N.[2,3] The weakest ceramic choice, a nonadhesively cemented porcelain jacket crown, has an adequate resistance to direct loading (545

N) for the anterior region. IPS Empress (Ivoclar Vivadent; 892 N) and IPS e.max (Ivoclar Vivadent; 1,200 N) have higher resistance to forces, which may be as much as an order of magnitude greater than those generated during normal function.[4,5] In addition, the ultimate strengths for axial and oblique loads on anterior ceramic systems exceed the maximum normal bite-force peaks.[4,5] Normal functional loads, therefore, should not damage all-ceramic crowns (Table 18-1). A systematic review showed survival rates of all-ceramic systems over 5 years to be equivalent to those of metal-ceramic systems. Lower survival rates for all-ceramic crowns are found when the crowns are placed on molars, where loads are significantly higher.[8]

| Table 18-1 | **Comparison of all-ceramic restorative systems[6,7]** | | | |
|---|---|---|---|---|
| **Product** | **Flexural strength (MPa)** | **Abrasiveness vs natural tooth** | **Special equipment** | **Other characteristics** |
| Traditional feldspathic porcelain | 110–150 | Varies; higher leucite content yields higher wear | Special refractory die | No core material; uniform translucency and shade throughout; etchable for bonding to tooth |
| **Pressable ceramics** | | | | |
| IPS Empress | 160–182 | Comparable with natural tooth except when lay ered with conventional feldspathic porcelain | Special oven, die material, and molding process | Etchable for bonding to tooth; core material shaded and translucent |
| Optimal Pressable Ceramic (Pentron) | 165 | Same as for IPS Empress | Same as for IPS Empress | Same as for IPS Empress |
| IPS e.max Press (Ivoclar Vivadent) | 400 | Less wear than natural tooth | Same as for IPS Empress | Crown can be cemented or etched to bond; possible to do three-unit fixed partial denture for anterior teeth |
| **Milled ceramics** | | | | |
| | | Less wear than | Special die scan ner; computer | Dense translucent core; not |

| | | | | |
|---|---|---|---|---|
| Procera (Nobel Biocare) | 600 | natural tooth | with modem; CAD/CAM machine | etchable, must be cemented |
| Zirconium oxide core ceramics | 1,000 | Less wear than natural tooth | Special CAD/CAM machine or dry pressed | Dense opaque core with limited dentin shades; not etchable, must be cement ed; may be used for limited fixed partial dentures |
| IPS e.max CAD (Ivoclar Vivadent) | 360 | Less wear than natural tooth | Special CAD/CAM machine and special oven for crystallizing the milled material | Milled material requires crystallization by heating at high temperature |

All-ceramic systems are not always more esthetic. In the hands of an average laboratory technician, all-ceramic systems will be repeatedly more esthetic simply because of the lack of a metal core[6,9] (Fig 18-1). However, given proper preparation depths, skilled technicians can mimic natural teeth using a metal-ceramic system (Fig 18-2). The anterior restoration being fabricated will also play an important role in the patient perception. In a survey, more patients felt that single all-ceramic crowns were more natural and preferred them over metal-ceramic crowns. However, the opposite was true of anterior fixed partial dentures, where metal-ceramic was deemed to be more esthetic and preferred.[10]



**Fig 18-1** All-ceramic crowns have a porcelain core, which facilitates an esthetic match. These crowns provide adequate strength to resist functional loads.




**Fig 18-2***(a)* Porcelain layered over the metal substructure. Similar layering is accomplished for zirconia cores to mask the white opaque core. *(b)* Existing metal-ceramic crowns appear monochromatic and opaque. *(c)* With proper tooth reduction, a master ceramist can give metal-ceramic crowns a very lifelike quality. (Crowns fabricated by Lebeau Dental Laboratory, Renton, Washington.)

Metal-ceramic crowns are more appropriate for some patients and restorations than others. Teeth that are opaque or have high chroma and high value may be easier to match with metal-ceramic systems than with all-ceramic systems. A patient with a low lip line is a good candidate for metal-ceramics, because the esthetic weakness of metal-ceramics is most pronounced in the gingival third, where the opacity influences the brightness. The highly reflective opaque ceramic coating used to mask the metal is difficult to disguise when matching teeth with low color saturation or brightness (Fig 18-3). In the anterior region, many patients demand a perfect match of crown, tooth, and gingiva. This can be challenging for metal-core crowns. In an attempt to eliminate the gingival opacity, some ceramists stop the extension of the metal substructure up to 2 mm away from the shoulder.[11] This removes the highly reflective opacity in the gingival third, decreasing reflection and allowing light penetration into the cervical tooth structure, without decreasing the strength of the crown.[12,13] Light transmission illuminates the gingiva and eliminates the dark gingival shadow sometimes found around metal-ceramic crowns.[14]



**Fig 18-3** Inadequate space for metal and porcelain may create unnatural opacity, as seen in this maxillary lateral incisor.

If practitioners should not make decisions based solely on the strength of metal-ceramic restorations or the esthetics of all-ceramic systems, then what criteria are

appropriate? When esthetic demands are high, an enamel-bonded veneer is the first choice. Adequate enamel must be present, and the patient should have no more than moderate parafunctional habits. It is commonly agreed that enamel-bonded veneers are the most predictable, most esthetic, and strongest restorations for anterior teeth (see chapter 17). However, veneers are contraindicated when a minority of the prepared tooth is in enamel; a sound bonding surface is critical to the strength and success of porcelain laminate restorations.

When veneers are contraindicated, choosing the correct anterior ceramic crown system is critical. The choice involves a hierarchy based on the esthetic goals of the patient, the preoperative condition of the tooth, and the load that the tooth will receive. The questions to be asked when deciding on the correct ceramic crown system are: *(1)* What are the esthetic demands, especially in the gingival third? *(2)* What is the quality of the bonding surface? *(3)* Does the patient have parafunctional habits? If so, what is the severity?

## Esthetic demands

An evaluation of esthetics and a diagnosis should be conducted following the guidelines established in chapter 3. Patient expectations should be addressed before treatment begins. If there are limitations in the treatment, they should be discussed in advance. Diagnostic wax-ups or computer imaging may help determine what can be achieved with the restorations and aid in patient communication. Elements that affect the esthetic choice of materials include the color of the underlying structures (eg, post and core, discolored dentin, implant abutment) and the importance of the esthetic match in the gingival third.[14]

Veneers bonded to enamel provide some of the most beautiful and dependable restorations for anterior teeth. The thin laminate of porcelain provides an optical refractive index similar to that of translucent enamel, allowing the natural tooth to act as the color substrate. Therefore, when the tooth substrate is an ideal color, a veneer restoration can be placed that is almost imperceptible (Fig 18-4).




**Fig 18-4***(a)* An increase in length and alteration of shape is required esthetically. The tooth color is acceptable, there are no restorations, and there is adequate enamel for bonding. Enamel-bonded

veneers are the restorations of choice. *(b)* When properly planned and constructed, enamel-bonded veneers are almost imperceptible.

A ceramic crown can be thought of as a veneer of "enamel" porcelain over a "dentin" ceramic core (Fig 18-5). This "dentin" core material can be feldspathic porcelain, castable glass, heat-pressed or milled leucite-reinforced ceramic, infiltrated alumina, heat-pressed or milled lithium disilicate, or zirconium oxide.[15] All cores can be laminated with veneering porcelain or stained. The type of core material used in a particular patient depends on the underlying tooth structure[16] (Table 18-2).



**Fig 18-5** Unlike metal-ceramic crowns, cemented all-ceramic restorations are significantly influenced by the underlying color. In this case, an amalgam buildup shows through an all-ceramic crown on the maxillary left central incisor, making it too gray.

Table 18-2 **Selection criteria for anterior ceramic crown materials[16]**

| Shade and appearance of natural teeth | Conventional feldspathic | Veneered OPC | Colored OPC | Veneered Empress | Colored Empress | IPS e.max | Procera | Zirconiu oxide |
|---|---|---|---|---|---|---|---|---|
| Vita A-1 to A-2; low color content, opaque, high brightness | × | | | | | | × | × |
| Vita A-3; moderate color content, translucent body, opaque body | × | × | × | × | × | × | × | × |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Vita A-3.5 to A-4; high color content, translucent or opaque | × | × | × | × | × | × | × | × |
| Altering shade from high color to low color | | | × | | × | × | × | × |
| Translucent, low brightness, high color | | × | × | × | × | × | | |
| Translucent, grayish teeth | | × | | × | | | | |
| Translucent, moderately bright teeth | × | | × | × | × | × | | |

Because the core material has a perceptible effect on crown color,[14,16] an ideal core material should match the natural optical properties of dentin and mask any discoloration present. Today, the feldspathic porcelains most closely mimic the opalescent and fluorescent properties of natural teeth. They are translucent, color stable, brilliant, and natural. Therefore, if the underlying tooth color is acceptable, full feldspathic crowns produce the most natural result, because they allow the underlying tooth color to show through.[17]

If the "dentin" core color must be altered, the ceramic core selection changes (Fig 18-6). IPS Empress, IPS e.max Ceram (Ivoclar Vivadent), Optimal Pressable Ceramic (OPC) (Jeneric/ Pentron), and Finesse All-Ceramic (Dentsply Ceramco) allow broader choices of substrate color with intermediate translucency. In-Ceram Alumina (Vident), Procera (Nobel Biocare), Lava Ceram (3M ESPE) and In-Ceram Zirconia (Vident) provide a high-strength core that is relatively opaque. The influence of the core material is most noticeable in the gingival third. The more translucent the core, the more gray it appears as the darkness of the oral cavity shows through. This is most evident with translucent OPC and IPS Empress. A higher value or brightness is produced by the more opaque, reflective cores found in

alumina, zirconia, and metal-ceramic crowns.[18]



**Fig 18-6** Dark prepared teeth are an esthetic challenge for all-ceramic restorations. The correct ceramic core must be chosen to minimize the effect of the discoloration.

## Quality of the bonding surface

Anterior crowns can be fixed to the tooth by traditional cementation or by bonding. Traditional cementation utilizes a cement such as glass ionomer or resin-modified glass ionomer. Bonding uses a luting resin or a resin composite and resin adhesive. (In this chapter, the terms *cement* and *cementation* will be used to discuss insertion with traditional cements. The terms *resin cement*, *self-adhesive resin*, *bonding resin*, and *bonding* will be used in reference to resin bonding of restorations.) A traditional cement occupies the space between the restoration and the tooth surfaces, and none but the glass-ionomer cements provide adhesion to tooth structure. In most cases, bonding provides adhesion to both surfaces. Because of this, a bonded crown may not have the same requirements for tooth preparation as a cemented crown. In addition, bonding acts to transfer force from the ceramic material to the underlying tooth structure and strengthens an all-ceramic restoration that could be relatively weak if cemented with conventional cements that are brittle and do not reinforce the intaglio surface as well as resins.

When most all-ceramic crowns are bonded, the ceramic is etched to create micromechanical retention for the resin cement. Bonding procedures, such as those described in chapter 9, are performed on the tooth. When the resin cement is polymerized, it forms a rigid union between the restoration and the tooth. Under function, the porcelain-resin-dentin bond allows the resin to transfer load to underlying tooth structure.[19] Properly bonded crowns have fracture strengths far greater than human bite force,[20,21] but functional and parafunctional forces and the hydrodynamic nature of dentin may decrease bond strength with time. If the dentin bond is compromised, the porcelain becomes more susceptible to fracture. A study by Neiva et al[22] comparing the fracture strength of three ceramic materials reported

that bonded IPS Empress had a higher mean fracture strength than In-Ceram Alumina or Procera. However, with a compromised bond, IPS Empress was significantly weaker than the other systems.

The nature of the dentin surface significantly impacts the strength of the tooth-adhesive interface (Fig 18-7). The risk of debonding and fracture is magnified when the underlying dentin is less than optimal. Clinicians must recognize differences in dentin composition before planning restorations that depend on long-term dentin bonding.[23] Clinical evidence suggests that loss of cervical bonded restorations is more prevalent with deep or sclerotic dentin. Changes in the microstructure of dentin due to acid degradation (resulting from bulimia and gastrointestinal reflux) yields an accelerated sclerosing of the dentin and poor bond strengths.[24] Deep dentin can also significantly affect strength at the dentin-resin interface.[25] The density of the dentinal tubules is greatest near the pulp. Tubules represent 28% by volume of the dentin along the pulpal wall, compared with 4% by volume at the dentinoenamel junction.[23] Adhesive bond strengths in deep dentin are generally lower because there is less intertubular dentin space for bonding and greater moisture flow that may interfere with bonding.[26]



**Fig 18-7** This patient displays signs of attrition from parafunc-tion, erosion from chewing aspirin and swishing acidic juices, and cervical notching from abrasion and abfraction.

In addition to a compromised dentin surface, bending forces beyond the ceramic limits are a contraindication to bonded ceramics. *Flexural fatigue*, or *abfraction*, is the pathologic loss of hard tooth substance by biomechanical loading forces (see Fig 18-7). This loss is thought to be the result of a combination of toothbrush abrasion and flexure and ultimate fatigue of enamel and/or dentin at some location distant from the actual point of loading and is more common in maxillary anterior teeth because of the higher buccal/lingual stress profile in the cervical region.[27] Eccentric forces can generate cervical flexure, resulting in stress concentrations in a bonded crown.[28] Cervical restoration failure occurs more commonly in mandibular incisors

than in maxillary incisors.[29] Tooth flexure can cause porcelain fracture or less obvious failure because of marginal leakage.

Carious or noncarious lesions or previous restorations may require a crown margin to be placed far apical to the gingival crest. Adhesion is affected by dentin moisture or other contaminants (blood, sulcular fluid, or saliva).[30–32] The inability to isolate subgingival margins prevents predictable dentin bonding and is therefore a contraindication to bonded ceramic crowns.[29] Strength is not a key criterion for a ceramic crown when the dentin bonding is maximized. When achieving the ultimate bond is compromised, however, a bonded ceramic crown may not be the restoration of choice.

## Parafunction

Parafunction is a physiologically normal activation of voluntary skeletal muscles to produce behaviors that lack functional purpose and are potentially injurious. It occurs cyclically, and because it is mediated by the limbic system, it is harder to control and change.[33] Human bite force on anterior teeth rarely exceeds 200 N, well within the tolerances of most ceramic systems. Forces generated during diurnal parafunctional activity may be of significant force and duration to be damaging to all-ceramic systems. This can result in sudden, catastrophic failure.

A more gradual failure of an all-ceramic system comes from fatigue failure. Testing confirms that cyclic loading is much more damaging than static loading, given the same total time and maximum load.[34] Cyclic loading in a moist environment significantly decreases the mean failure load of an all-ceramic system.[35] Three modes of cyclic fracture can occur in an all-ceramic crown near the area of contact with an opposing tooth: *(1)* brittle-mode tensile cracks beginning on the occluding surface, *(2)* quasi-plastic mode microfractures within the ceramic material in a "yield" zone, and *(3)* radial cracking (undersurface tension flexes the core/ceramic veneer interface).[36] It has been postulated that these inner-surface cracks are the most likely cause of failure of cemented all-ceramic systems.[37] All of the fatigue modes of failure can occur with repeated functional loads. All-ceramic crowns have been shown in laboratory studies to be more susceptible to fatigue than metal-ceramic crowns.[38] However, severe loading during parafunctional episodes magnifies the damage by coalescence of microcracks into a macroflaw, accelerating the loss of strength and reducing the life span of the ceramic restoration. A thorough evaluation of the patient's parafunctional history is required in choosing an anterior ceramic system.

# Hierarchical Approach

Once practitioners have answered questions regarding esthetics, bonding substrate, and parafunction, a hierarchical approach can be used to select the best anterior ceramic restoration for each patient. The choices begin with veneer crowns and conclude with metal-ceramic crowns. The type of ceramic crown is chosen when a tooth meets all the indications for that level. Refinement within the hierarchy is then made by meeting additional criteria.

## Veneer crowns

*Indications*

1. Esthetics of primary importance
2. Mixed enamel and dentinal substrate
3. Minimal or no parafunction

Enamel-bonded porcelain laminate veneers are the most dependable and esthetic anterior ceramic restorations. A veneer allows the passage of light into the tooth, and the underlying tooth color provides a natural effect. A *veneer crown* is simply a veneer that covers the entire tooth. It is to be used only in selected cases, because transitions from thin to thicker porcelain make fabrication difficult. The preparation preserves remaining enamel and uses a conservative preparation design. The most common indication is for a peg-shaped lateral incisor. Another indication is a tooth with good enamel support, large proximal restorations, and endodontic access (Fig 18-8). The enamel is reduced by 0.5 mm, the proximal restorations are removed, and the margins are finished with a light chamfer.



**Fig 18-8***(a)* This lateral incisor had a large area of enamel available for bonding after preparation, but the interproximal restoration and lingual endodontic access made it an ideal candidate for a veneer crown. *(b)* The preparation displays a subtle transition from 0.5 mm of facial reduction to 1.0 mm of lingual reduction. Because of caries gingival to the lingual composite, the lingual preparation extends into dentin in this one area. However, all other areas of the preparation are in enamel, including the cavosurface margins. *(c)* The veneer crown is in esthetic harmony with the natural central incisor and veneered canine.

## Bonded all-ceramic crowns

*Indications*

1. High esthetic demands, especially in the gingival third
2. Dentin bonding substrate acceptable
3. Moderate to no parafunction

Key to bonded all-ceramic restorations is the ability to provide predictable dentin adhesion. Without a sound dentin-crown unit after bonding, all-ceramic bonded crowns are extremely weak and prone to fracture.[18–21] This class of ceramics includes feldspathic porcelain (on a refractory die or platinum foil), castable glass, injection-molded ceramics, and computer-aided design/computer-assisted manufacture (CAD/ CAM) milled crowns. These systems provide the finest esthetics of the anterior ceramic crown categories because they most closely match the translucency, brilliance, and qualities of natural tooth substrates. The difference between the systems is the varying degree of translucency in the core materials. Choices within this category should be based on which substrate most closely mimics that of the natural teeth. If the prepared tooth color is normal (Fig 18-9a), full feldspathic crowns produce the most natural core and have a biologic outcome and survival probability comparable with cored ceramic systems[39] (Fig 18-9b). However, when the crown substrate requires alteration of translucency or opacity, the core choice should be made accordingly.




**Fig 18-9***(a)* When the tooth substrate is acceptable for bonding and color, a bonded, feldspathic all-ceramic crown is indicated. *(b)* The bonded feldspathic crown provides functional predictability and natural esthetics.

IPS Empress and OPC are both heat-processed or computer-milled, high-leucite pressed ceramics. They provide laboratories with multiple dentin shades and translucencies, which can be used to mask defects or match adjacent dentition. Once the proper dentin shade, or *stump*, is chosen, the core is fabricated using a lost-wax technique. The ceramist decides whether the crown will be cast in its entirety and then colored with surface stains (the *monolithic technique*) (Fig 18-10) or a cast

coping will be layered with feldspathic porcelain (the *veneered technique*) (Fig 18-11). Surface staining involves placing heavily pigmented characterization colors on the crown, which is then covered with a glazing powder and fired. The veneered technique is the recommended method for esthetic anterior ceramics, and the monolithic technique is the recommended method for strength (Fig 18-12). When the veneered technique is used, the preparation depth must allow adequate space to produce a substructure that resembles a veneer preparation. Enamel and incisal feldspathic porcelains are then veneered on the core. Both techniques provide very promising clinical results.[40] IPS Empress and OPC have comparable flexural strength.[41] The in vitro marginal fit of IPS Empress has been reported to provide a mean marginal opening of 28 to 99 µm, with the type of finish line explaining the difference in marginal fit.[42] A survival rate for anterior crowns of greater than 95% has been reported through 11 years of observation.[40] Heat-pressed ceramics offer mixed reports on wear values comparable with those of enamel.[43,44] Routine monitoring of opposing tooth wear is therefore important.

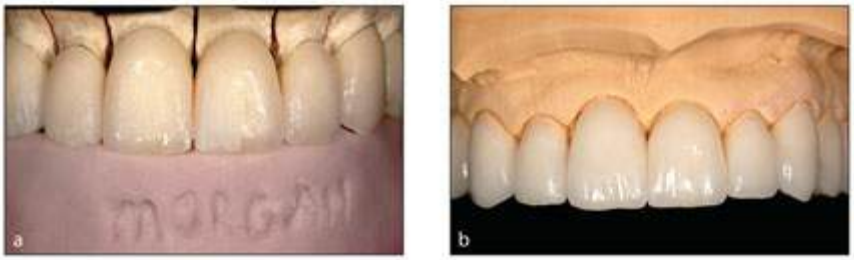


**Fig 18-10***(a)* Monolithic ceramic restorations with the addition of characterizing staining effects. *(b)* Completed monolithic ceramic restorations with stain.



**Fig 18-11***(a)* Wax-up with cutback allowing room for veneering ceramic. Note the design of the cutback to maintain the strength of the incisal edge in core ceramic. *(b)* Lithium disilicate pressed to allow space for veneering ceramic. *(c)* IPS e.max veneered definitive restorations.



**Fig 18-12** The maxillary central and lateral incisors are restored with veneered IPS Empress crowns. When the tooth color is poor but a reliable bond is available, IPS Empress is a good choice because several core colors are available.

## Cemented all-ceramic crowns

*Indications*

1. Moderate to high esthetic demands, especially in the gingival third
2. Poor-quality dentin bonding substrate
3. Moderate to no parafunction

Cemented all-ceramic crowns are used when esthetics is an overriding concern but the dentinal substrate does not provide for proper bonding. Because the core material is very strong, bonding to the underlying tooth structure is not necessary. Cementation requires fewer steps, is less technique sensitive, and has less opportunity for mishaps than do bonding procedures. Because the crown is cemented, the preparation must meet the retention and resistance requirements of any cemented crown. Esthetics is compromised slightly because the ceramic core is less translucent than bonded ceramic restorations and is more opaque than many natural tooth substrates. Because of the high strength of these ceramic materials, cemented ceramic crowns can be placed in patients with moderate or controlled parafunction.

The difference between the ceramic materials that must be bonded versus those that can be used in cemented crowns is found in the composition and processing techniques for the core. For example, In-Ceram Alumina is an 85% alumina, glass-infiltrated core. In-Ceram Spinell (Vident) is a mixture of alumina and magnesia made on a resin die. Procera crowns have a 99% alumina core fabricated on a die designed from digitized specifications made from the master die. IPS e.max is a lithium disilicate–based glass ceramic. In-Ceram Zirconia is a slip-cast or dry-pressed zirconia core.

The In-Ceram Alumina core may be fabricated through CAD/CAM milling or slip casting. Slip castings are created on a special gypsum die to which an alumina and water mixture or “slip” is applied and shaped. The core is sintered (baked at high

temperature) in a furnace, creating an interconnected porous network. The core material is then returned to the porcelain oven, and a lanthanum aluminosilicate glass is infiltrated into the pores of the core to add strength. The In-Ceram Alumina slip casting technique has been used for well over a decade. Today, milling technology has allowed crowns to be produced from a ceramic block with greater flexural strength than slip casting. Flexural strengths from 300 to 600 MPa have been reported for In-Ceram Alumina.[45] A clinical study of 163 anterior units demonstrated a 15-year survival rate of 87.5%, which is comparable with reports on metal-ceramic crowns.[46] Data from a different study showed that core fractures and porcelain fractures occurred at a rate of approximately 0.6% and 0.3% per year, respectively.[47] Additional in vivo studies confirm the high success rate of In-Ceram Alumina.[48,49] Marginal integrity has been shown to be within clinically acceptable standards.[50]

Procera is another system that uses an alumina core (Fig 18-13). The master die is scanned into a computer, and the surface contour of the die is mapped with the use of more than 50,000 measured values. An alumina coping is designed on the computer, and the relief space for the cementing agent is established. The data are then transmitted via modem to a production station, where the coping is manufactured with advanced powder technology and a CAD/CAM technique. The coping contains high-purity (99.9%) aluminum oxide powder, which is milled and sintered. A veneering porcelain that is compatible with the coping is layered to develop crown contours and esthetics. The flexural strength of the core is reported to be greater than 600 MPa, with a fracture toughness of 4.48 MPa-m^0.5.[51] The mean load to failure of the core/veneer porcelain system is clinically acceptable when the veneer porcelain is more than 0.4 mm thick. The veneering porcelain doubles in strength at 0.9 mm.[52] With a 0.6-mm-thick Procera core, a tooth reduction of 1.0 to 1.5 mm is required. In a laboratory study, mean gap dimensions for marginal openings, internal adaptation, and precision of fit for Procera crowns were less than 70 µm.[53] Data on the clinical fit of Procera crowns indicated a mean marginal gap width of 50 to 135 µm for crowns depending on the finish line configuration.[54] The survival rates of Pro-cera crowns were reported to be 97% to 100% in the anterior region after 5 and 7 years regardless of the cement used.[55,56]



**Fig 18-13***(a)* The post and core does not allow bonding. A cemented crown is an appropriate choice for the central incisor. *(b)* A Procera crown was chosen to mask the color of the post and core and to esthetically match the other all-ceramic restorations.

Lithium discilicate provides clinicians with an extremely strong, etchable core material, making it bondable or cementable. This allows IPS e.max to be used for almost any fixed prosthetic need: veneers, crowns, and fixed partial dentures (Fig 18-14). Because of its ability to be cemented, IPS e.max is placed in the cemented crown category for this evaluation. The material is currently available in two varieties: IPS e.max Press or IPS e.max CAD. The pressed option requires the core pattern to be waxed to full contour and invested. The IPS e.max glass-ceramic ingots are heat pressed in a 920°C furnace under pressure and vacuum. The glass ceramic becomes viscous and flows into the lost-wax mold of the core. The IPS e.max core is coated with a glass ceramic containing fluoroapatite. These apatite crystals are similar in structure to those found in natural teeth. Reportedly, they allow the porcelain to mimic the optical properties of translucency, brightness, and opalescence of a natural



**Fig 18-14***(a)* Multiple dentin shades and poor bonding substrates are indications for IPS e.max crowns. *(b)* The system allows ceramists to mask the tooth discoloration and provides an opportunity for the dentist to bond or lute the crowns. *(c)* The IPS e.max crown on the left central incisor provides natural esthetics even over an implant abutment.

tooth.[57] The IPS e.max CAD "blue block" is a lithium metasilicate crystal milling block. A milling unit creates a crown from a pattern generated in a computer. At this stage the crown is very weak. A second heat-treating process is performed in a porcelain furnace at approximately 850°C; the metasilicate is dissolved, and the lithium disilicate crystallizes. This results in a fine-grained glass ceramic with 70%

crystal volume incorporated into a glass matrix.[58] The IPS e.max core has a flexural strength of 350 ± 50 MPa, a fracture toughness of 3.2 MPa, and a failure load of 771 to 1,115 N.[59,60] In 2- to 5-year evaluations of IPS e.max restorations placed in general practice and in controlled clinical settings, the restoration survival rate was 94% to 100%.[61–64] Results from a study on the mean gap dimensions and marginal opening for incisor crowns showed that IPS e.max restorations had the smallest and most homogenous gap dimensions (46 ± 1 μm) compared with milled and infiltrated zirconia crowns.[65] In addition, two studies reported no evidence of abrasion of the ceramic or the opposing teeth in vitro or in vivo.[44,66]

All-ceramic cementable crown cores may also be fabricated in zirconium oxide. Numerous manufacturers provide zirconia cores with unique fabrication techniques and chemistries. Zir-conia has outstanding mechanical properties and high biocompatibility. Zirconium oxide produces an extremely strong core (998 to 1,183 N).[60,67,68] It is stronger than all other ceramic core materials.[69] Zirconia frameworks may be slip-cast like In-Ceram, dry-pressed, or milled from a solid block. Copings produced by a CAM system exhibited a mean marginal gap (66.4 ± 42.2 μm) well below the clinically acceptable 100-μm limit.[70] The relative translucency of a zirconium oxide ceramic core is roughly equivalent to that of a metal core.[18] Individual systems and framework thicknesses impact the levels of light transmission, making some systems more appropriate for anterior application and others for masking discoloration.[71] A recent comprehensive review of the literature on zirconia single crowns found limited long-term survival data.[72] A 3-year study of single units produced in a private practice showed a cumulative survival rate of 93%, with the majority of failures due to ceramic veneer chipping.[73] The drawback to the restoration continues to be the weaker veneering porcelains (Fig 18-15). Microtensile bond strengths vary significantly between zirconia substrates and veneering ceramics. Failures are mainly interfacial.[74] The rate of veneering ceramic fracture is significantly higher than with metal-ceramic restorations.[75] New approaches to zirconia coping design and veneering ceramic cooling rates have begun to improve the survival rates.[76] In an attempt to limit failures, researchers have focused on utilizing monolithic zirconia restorations. This is especially popular for posterior crowns and fixed partial dentures. The use of preshaded zirconia blocks, the use of a dye to shade a partially sintered zirconia block, and external shading and glazing have provided durable crowns but only acceptable esthetics for anterior restorations.[77]



**Fig 18-15** Chipping of veneering ceramic on zirconia crowns is a significant disadvantage.

## Metal-ceramic crowns

*Indications*

1. Moderate esthetic demands
2. Poor-quality dentinal substrate
3. Severe to no parafunction

Porcelain-fused-to-metal crowns have a long history of success.[78] Introduced in the late 1940s, metal-ceramic crowns continue to be the most common complete-coverage anterior restoration. Porcelain margins were developed in the 1960s, and the technique later evolved with the introduction of shoulder porcelains.[13] Recent development of opalescent and fluorescent porcelains has dramatically improved the esthetics of these restorations.[79]

The quality of the dentinal substrate does not play a role in the choice of metal-ceramic restorations because they are cemented, but resistance and retention requirements are of major importance. Patients with moderate to severe parafunction are best treated with porcelain-fused-to-metal crowns, which are inherently stronger than all-ceramic crowns.[80] Tooth contacts may be developed in metal rather than porcelain to decrease wear of the opposing dentition. Other indications for metal-ceramic crowns are a deep, tight incisor relationship and minimal dentin wall thickness after endodontic treatment. Neither situation allows sufficient reduction for all-ceramic crowns. Preparations can be modified because metal surfaces as thin as 0.5 mm are usually adequate (Fig 18-16). Metal ceramics are still preferred for fixed partial dentures even when patient perception of esthetics is key.[10]

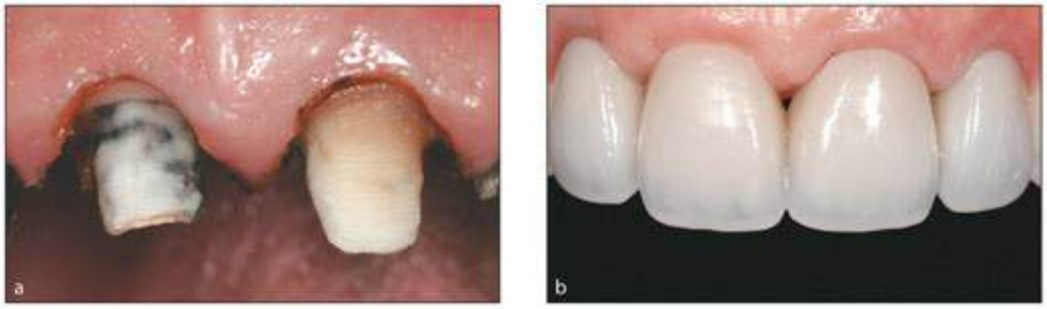


**Fig 18-16***(a)* Endodontically treated and structurally compromised teeth are indications for metal-ceramic crowns. *(b)* Metal-ceramic crowns restore the central and lateral incisors. A porcelain butt-joint margin and 2 mm of metal reduction from the facial shoulder allow light to enter the tooth and illuminate the gingival third of the crown.

Today, clinicians have a plethora of anterior ceramic systems from which to choose. The decision should be based on esthetic goals, bonding requirements, and parafunctional habits rather than a favorite ceramic system. Dentists using a hierarchical approach to selecting anterior ceramic crowns will discover that several ceramic materials may be used within the same smile (Fig 18-17).

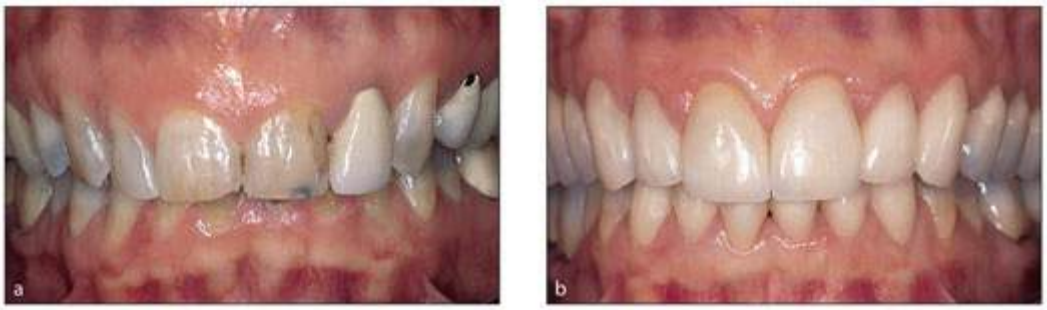


**Fig 18-17***(a)* A careful consideration of the esthetic demands, bonding substrate, and any parafunctional habits may lead to the use of multiple systems within the same patient. *(b)* Veneers and bonded and cemented crowns were used to restore function and esthetics.

## Anterior Tooth Preparation

To provide a tooth preparation that meets the restorative and technical guidelines and provides predictable results, practitioners must understand and blend the biologic, mechanical, and esthetic components of tooth preparation.[81] All three categories are equally important. Biologic principles dictate finish line position and pulpal preservation. Mechanical preparation principles include retention and resistance features, margin configuration, accuracy of fit, and integrity. Principles of esthetics require that the tooth-restoration interface is not a visual focal point.

# Biologic principles

## *Finish line position*

Margin location is the most important biologic parameter in predictably maintaining gingival health.[82–85] Even among patients receiving regular preventive dental care, subgingival margins are associated with increased probing depths and gingival inflammation.[82–84] The critical factor in maintaining healthy gingival tissue is the relationship between margin location and supracrestal fiber attachment.[86,87] If the restorative margins impinge on the supracrestal fiber attachment, chronic inflammation can result.[88–90] Practitioners must therefore balance the principles of periodontal health and the desire for concealed porcelain margins.[91] They must also be aware of the cervical limitations of crown preparations; a complete understanding of the dentogingival complex is paramount.

In 1921, Gottlieb[92] first described "an epithelial attachment of the gingiva to the hard tissues." Sicher[93] discussed the "dento-gingival junction" as the fibrous, connective tissue attachment of the gingiva and the epithelial attachment. The dimensions of this functional dentogingival unit were first reported by Gargiulo et al[94] (Fig 18-18a), who described the dentogingival junction as consisting of a sulcus, junctional epithelium, and gingival fiber attachment to cementum, coronal to the alveolar crest. The dimensions of the dentogingival junction were further refined by Vacek et al.[96] The term *biologic width* was given to this zone of connective tissue and epithelium by D. Walter Cohen (1962) in an unpublished presentation at Walter Reed Army Medical Center. This biologic zone was reintroduced to periodontics and restorative dentistry by Ing-ber et al[97] in 1977.

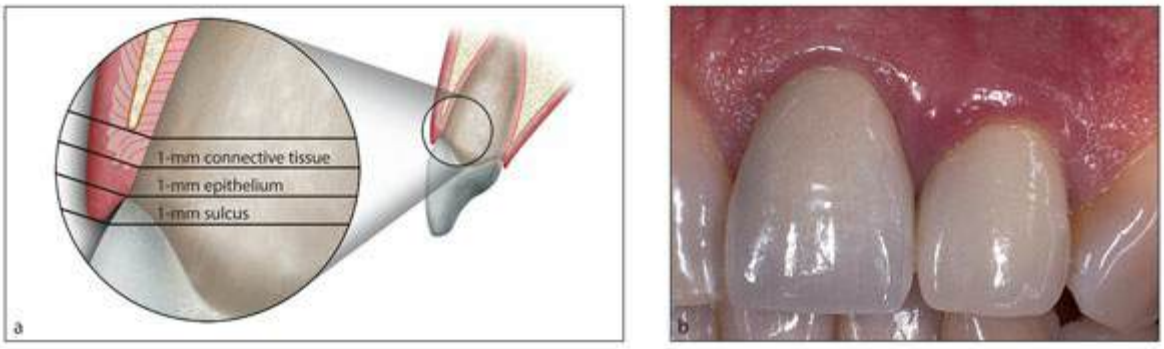



**Fig 18-18***(a)* The biologic width comprises epithelial and connective tissue attachments. Crown preparations that impinge on the biologic width result in chronic inflammation. (Reprinted from Robbins[95] with permission.) *(b)* This crown preparation impinged on the biologic width interproximally. Inflammation had been present for 17 years.

Vacek et al[96] provided an arithmetic outline for evaluating the tissue-tooth interface. The average for the connective tissue attachment, the least variable of the biologic components, is 0.71 mm (0.35 to 1.74 mm). Connective tissue includes 10 different gingival fiber bundles, the periodontal ligament, cementum, and collagen fiber bundles embedded into the root surface; these embedded fiber bundles are called *Sharpey's fibers*. The connective tissue zone also contains the nerve and blood supply for the gingival tissues. The connective tissue attachment is the strongest part of the attachment and provides resistance that, under normal circumstances, prevents a periodontal probe or gingival retraction cord from penetrating to the bone. It includes the circumferential fibers that often produce facially evident inflammatory reactions to interproximal biologic width violations.[89,90]

Coronal to the connective tissue attachment is the epithelial attachment. The epithelial attachment has an average dimension of 1.03 mm (0.38 to 2.48 mm). The hemidesmosomal attachment of the epithelium provides a tight approximation of tissue to the tooth but is easily penetrated, especially in the presence of inflammation. The cells in this layer undergo a continual coronal migration with a complete turnover every 4 to 6 days.[98] They attach to enamel, cementum, dentin, and even porcelain.

The term *biologic width* describes a vertical measurement of 1.74 mm, the combined width of the connective and epithelial attachments. If the margin of a restoration violates the 2-mm biologic width, substantial gingival inflammation often results[88–90] (Fig 18-18b).

The difficulty for practitioners is that Gargiulo et al and Vacek et al's work presents a "contrived illusion of mathematical precision." The research suggests a vertical measurement of 1.19 mm for the sulcus. This would make the total dentogingival complex, biologic width plus sulcus, approximately 3 mm. Yet, clinically, the dimensions vary greatly from 0.43 to 6.03 mm. Not every tooth has the average biologic width of 2 mm and a 1-mm sulcular depth.[97] Each tooth presents unique gingival measurements that must be assessed and used in treatment. Individual measurements of the total dentogingival complex must, therefore, be used in making restorative decisions.[99]

If the goal is to place a restorative margin in the sulcus without violating the biologic width, the base of the sulcus must be identified. However, this is extremely difficult. The periodontal literature indicates that the tip of the periodontal probe often penetrates the base of the sulcus and may extend into the connective tissue.[100]

The depth of penetration depends on the level of inflammation, the diameter of the probe, and the pressure used on the probe. Because the sulcus depth can be identified only histologically, the distance from the free gingival margin to the crest of the alveolar bone is the only predictable measurement available to determine intracrevicular margin location. At the crown preparation appointment, the entire dentogingival complex is measured. After the administration of local anesthesia, a periodontal probe is pushed through the sul-cus until resistance is felt (Fig 18-19a). The probe is then angled away from the clinical crown while the tip is still touching the root surface (Fig 18-19b) and is pushed completely to the osseous crest (Fig 18-19c). This process is called *bone sounding*, and measurements are taken on the midfacial aspect of the tooth and at both facioproximal line angles. If the probe is not angled correctly, there is a greater risk of the tip of the probe skipping past the thin facial plate of bone, resulting in an inaccurate measurement.




**Fig 18-19***(a)* Crest classifications are based on the probing depths on the facial and interproximal surfaces from bone to free gingival margin. After the administration of local anesthesia, the probe is placed in the sulcus and pushed until it meets resistance. *(b)* The probe is angled to keep its tip on the root surface. *(c)* The probe is forced through the attachment apparatus until it engages bone. The depth is measured at the free gingival margin and the crest type classified.

Measurements on anterior teeth can be categorized into three types of relationships between the free gingival margin and the alveolar crest: normal, low, and high.[99] This relationship will influence margin placement, determine the stability of the attachment levels of the gingiva against the tooth, and influence the need for crown lengthening surgery prior to restorative procedures. The critical factor in proper management of the soft tissues is accurate location of the alveolar crest, which allows the clinician to avoid impingement on the biologic width.

**Normal crest relationship.** In a normal crest relationship, the measurement from the free gingival margin to the osseous crest is 3.0 mm facially and 3.0 to 4.5 mm interproximally (Fig 18-20), which usually results in a gingival scallop of 3.0 to 4.0 mm and tissue levels that are stable in relation to the tooth. The normal crest is found

on 85% of anterior teeth.[99] In the normal crest relationship, restorative margins can be placed 0.5 to 1.0 mm into the sulcus facially and 0.5 to 2.5 mm interproximally. The apical limit of the restorative margin is 2.5 mm coronal to the osseous crest. The retraction technique is not critical in this crest relationship because the gingival level is stable. Typically, a normal crest relationship should yield no recession and no loss of papilla height following routine intervention. Research indicates that a normal crest relationship will reestablish itself even if the tissue is completely denuded, although it may take up to 3 years to return to its normal form.[101,102]

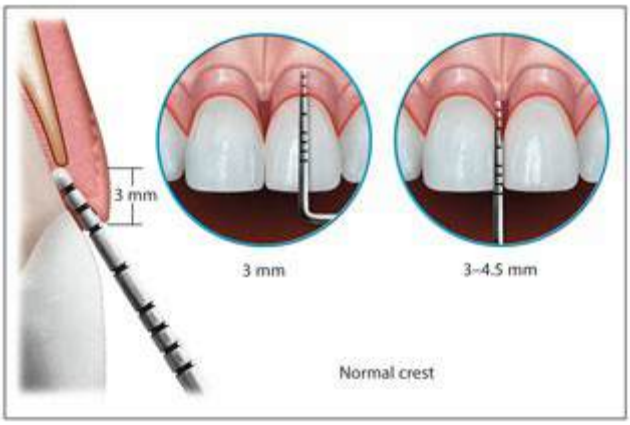



**Fig 18-20** Bone sounding depths of 3.0 mm facially and 3.0 to 4.5 mm interproximally represent a normal crest relationship. The tissue should rebound completely after manipulation.

**Low crest relationship.** A low crest relationship is the most difficult of all crest positions to manage and is found in 13% of anterior teeth.[99] The relationship of the free gingival margin to the osseous crest is greater than 3.0 mm facially and greater than 4.5 mm interproximally (Fig 18-21). The gingival scallop does not mimic the osseous crest. Patients with low crests are at high risk for facial recession and loss of papilla height because of the increased distance from the alveolar crest to the gingival margin. The position of the soft tissues on the tooth is not stable in teeth with a low crest relationship and can easily be altered unintentionally during treatment.

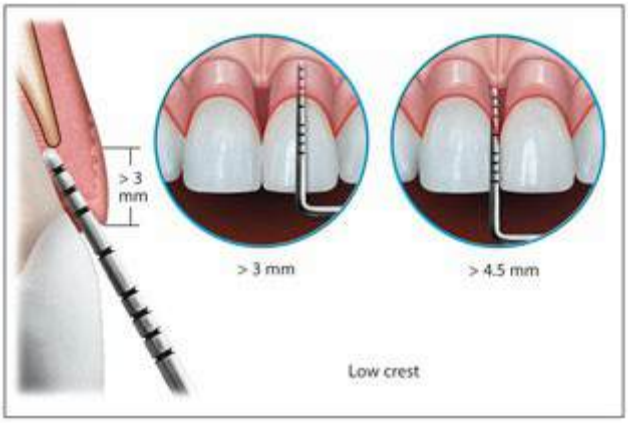



**Fig 18-21** Bone sounding depths of greater than 3.0 mm facially and 4.5 mm interproximally represent a low crest relationship. Tissue response is not predictable after manipulation. Recession and "black triangles" are probable.

If maintenance of the tissue levels is critical during restorative procedures, practitioners have two options.[95] One option is to correct the low crest surgically before tooth preparation, creating a normal crest relationship and thus achieving predictability. This can be accomplished by reducing the tissue height with an internally beveled gingivectomy so that the gingival crest is 3 mm coronal to the osseous crest. However, if the position of the cementoenamel junction, root anatomy, gingival architecture, osseous support, or esthetic demands prevent proactive treatment, the second option is to take great care to avoid damage to the attachment during preparation and impression making. The finish line of the preparation should be located at or coronal to the free gingival margin, and there should be minimal, if any, tissue retraction during impression making. The patient should be warned of the possible tissue changes before the preparation begins and have an understanding of the treatment options if tissue loss does occur. While not predictable, thicker tissue seems more resistant to recession following intervention.[103]

**High crest relationship.** Patients with high crests are the least common (2%) and pose the greatest risk for violation of biologic width.[99] Probing measurements are less than 3 mm facially or interproximally (Fig 18-22). The tissue levels are very stable, and the gingival scallop is flat, less than 3 mm. The high crest relationship sometimes occurs when excessive tissue covers the anatomical crown, such as in altered passive eruption[104] or in patients with noncarious cervical lesions. However, it is most common adjacent to edentulous spaces where the gingival scallop has flattened. Margin location is determined by the demands of biologic width. High-crest teeth, by definition, will only allow an equigingival or supragingival restorative margin because of the short distance to the alveolar crest.

These teeth are at high risk for biologic width impingement with intracre-vicular margin placement. Gingival retraction for impressions should be minimal.

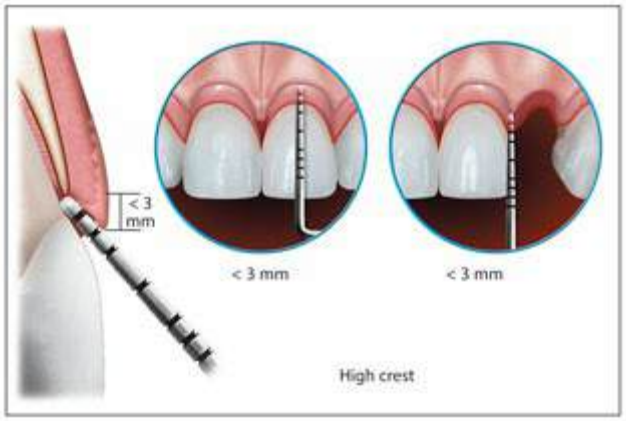



**Fig 18-22** Bone sounding depths of less than 3.0 mm facially and interproximally represent a high crest relationship. Intracrevicular preparation is difficult without biologic width violation.

### *Pulpal preservation*

Preoperative radiographs and pulp testing are important steps in determining pulp vitality prior to tooth preparation. Unfortunately, pulp testing cannot identify degrees of health. Separately or cumulatively, the effects of large restorations, leaking restorations, caries lesions, deep cracks, pins, etc, increase the chances of pulpal necrosis after tooth preparation. In one study, irreversible pulpitis occurred in 5.7% of cases in which crowns were placed on vital teeth.[105] Patients should be made aware of this risk preoperatively.

If the tooth preparation involves an increase of heat to the tooth, pulpal necrosis can occur. In one in vivo study in primates, pulpal injury occurred in 15% of teeth with a 5.5°C rise in temperature. An 11.1°C rise led to necrosis of the pulp in 60% of the teeth, and a 16.6°C rise caused necrosis of all the teeth tested.[106] Temperature changes have been monitored during complete crown preparation. When an air-water spray coolant was used, a temperature decrease in the pulp chamber from 37°C to 25°C after 4 minutes of exposure occurred. However, when only air coolant was used, the pulpal temperature rose from 37°C to 48°C after 1 minute of continuous exposure.[107] Therefore, continuous air-water coolant is a critical factor in maintaining pulpal health.

## Mechanical principles

During tooth preparation, several mechanical principles must be followed. The preparations must incorporate retention and resistance form, structural durability,

and marginal integrity.

## *Resistance and retention*

*Retention* is the feature of a crown preparation that resists dislodgment in a vertical direction or along the path of placement. In 1926, Ward[108] became the first practitioner to establish a standard for preparation taper. He prescribed 5% to 20% per inch or 3 to 12 degrees. Since then, recommendations have ranged from 3 to 5 degrees, to 6 degrees, to 10 to 14 degrees.[109] Jørgensen[110] indicated that there is a 50% reduction in retention when going from 5 to 10 degrees of taper. These tenets are heavily based on clinical empiricism and on two experiments in which crown and abutment analogs were pulled apart along their paths of insertion. The theoretical benefits of preparations with minimal convergence angles do not withstand scrutiny,[111–113] and such preparations are difficult to produce in clinical practice. Divergence from parallel might have to be as much as 12 degrees to be observed and produced clinically.[61,114] Routine preparations in practice have been measured at between 15 and 36 degrees without apparent detrimental effect to the longevity of the restorations.[115,116] For nonadhesively cemented restorations, the minimum convergence value required clinically is unknown, although total convergence up to 20 degrees has been shown to be acceptable.[113,117] Today, crowns with greater taper may be cemented with adhesive resin cements, minimizing the need to prepare the crowns to minimal convergence angles.[118]

*Resistance* is the feature of a tooth preparation that enhances the restoration's stability and resists dislodgment along an axis other than the path of placement.[119] Most retention studies utilize conventional pull-type tests to evaluate preparations and/or cements.[120,121] However, data on functional force vectors in the oral environment strongly suggest that these lift-off type forces are virtually nonexistent in the mouth.[122–124] During chewing, teeth are subjected to alternating combinations of buccolingual and occlusogingival forces.[125] These studies indicate that stresses that cause failure of an anterior restoration are repeated perpendicular or oblique forces.[112,124] Therefore, as Caputo and Standlee[126] concluded, "Resistance form is the most important factor that must be designed into any restoration if it is to succeed in function."

Resistance clinically is multifactoral. It is based on preparation taper, height, diameter, and cement type. Crowns generally loosen and fail by cleavage of the cement attachment without damaging the abutment or restoration.[113] The cement attachment fails when a portion of the abutment subjected to compressive and shear

forces is unable to withstand load application. Attachment failure is a progressive phenomenon linked to increasing abutment taper. Increasing the preparation taper from 10 to 20 degrees creates a broader stress distribution and greater stress within the cement.[113] The stress fields are consistently higher with greater taper and less surface area. The more parallel the walls of a preparation, the less stress on the cement lute irrespective of the preparation height. Taper becomes particularly important in teeth with a lower preparation surface area, such as an anterior tooth.[127] Clinically, a minimal preparation taper decreases the damaging effects of occlusal stress on the cement attachment, improving a crown's resistance even more than auxiliary preparation features like grooves or boxes.[128]

The height and diameter of the final preparation are also related to resistance.[129] Resistance is increased by lengthening the axial walls of the preparation.[130] The minimum height for resistance is one-half the diameter of the tooth.[131] This means that, on average, an anterior preparation must be 3.5 mm and a posterior preparation 4.0 mm in height. Of the total preparation height, the gingival 2.0 mm of the preparation must be on sound tooth structure to provide a proper ferrule,[132–135] and the other 1.5 mm or more can be in either tooth structure or buildup material (Fig 18-23). A *ferrule* is the marginal band of a crown that contacts tooth structure, providing protection from masticatory forces. In addition, the ferrule requires a dentin thickness of 1 mm from the external surface of the crown preparation to the wall of any endodontic preparation. In a comparison of in vivo restorations with and without 2 mm of remaining coronal tooth structure, the failure rate of the < 2 mm group was four times that of the control (26.20% and 6.67%, respectively)[136] (Fig 18-24).




**Fig 18-23** The minimum core height for an anterior ceramic crown is 3.5 mm. The cervical 2.0 mm of the facial and lingual aspects must be solid tooth structure for a proper ferrule.



**Fig 18-24** Inadequate coronal preparation height significantly increases risk of failure.

Resistance is also affected by the mechanical properties of the cement.[113] The limiting threshold of each crown is the cement's resistance to fatigue under cyclic stress.[113] The more stress that will be placed on the cement, because of a severe taper and/or lack of preparation height, the more resistant the cement must be. Resin-modified glass-ionomer cements are more resistant than conventional glass-ionomer cements, which are in turn more resistant than zinc phosphate cements. Most research on resistance and retention data is conducted on nonadhesively cemented crowns. Today, however, adhesive cements allow the placement of crowns that do not meet standard taper and length requirements. Yet, because the hydrodynamic nature of dentin bonding makes it unpredictable, it is suggested that all crown preparations meet minimum requirements.[29]

## *Structural durability*

*Structural durability* is the relationship between occlusal stress and material strength. It ensures that a restoration does not deform or fracture under load. In a metal-ceramic crown, the minimum metal thickness under porcelain is 0.4 to 0.5 mm for gold alloys and 0.2 mm for base-metal alloys. If the metal is too thin, it will flex under load, resulting in possible porcelain fracture. The minimum porcelain thickness over metal is 0.9 mm (0.2 mm for the opaque material and 0.7 mm for body porcelain). Ceramists prefer a 1.3- to 1.5-mm reduction for the axial surfaces of metal-ceramic crowns and a 2.0-mm reduction incisally/occlusally. The greater the reduction, the easier it is to mask the opaque material in the gingival third of the crown with body porcelain. Most bonded all-ceramic crowns require a minimum thickness of 1.0 mm to provide esthetics and adequate strength. Cemented all-ceramic crowns require an average circumferential tooth reduction of 1.5 mm for strength.

## *Marginal integrity*

A completely closed margin is unattainable clinically. Even the finest margins are not sufficiently closed to prevent bacterial ingress. To place it in perspective, the width of a human hair is 50 μm; bacteria responsible for caries are 4 to 5 μm in diameter. Because bacteria are constantly passing under restoration margins, patient resistance to disease is more important than the marginal opening of crowns. What, then, is an acceptable marginal opening? One study reported that when the margin of an inlay or onlay could not be visualized, a marginal discrepancy of 119 μm was found to be acceptable.[137] Björn et al[138,139] reported that 83% of gold and 74% of porcelain crowns exhibited marginal defects; more than half were greater than 200 μm. However, defective margins are to blame in only 10% of failed restorations. While all practitioners should strive for the finest margins possible, it is impossible to achieve a closed margin. The best possible margin enables the patient to floss and care for the restoration, minimizing cement dissolution and maximizing the patient's natural resistance factors. Because 100 μm is the smallest detectable ledge,[140] this can be used as a practical criterion for evaluating fit.

What finish line design provides the best marginal integrity? Traditionally, the margin design selected does not have a significant effect on marginal seal of metal-ceramic restorations.[141,142] Shoulder (Fig 18-25a), shoulder-bevel (Fig 18-25b), and chamfer (Fig 18-25c) finish line preparations all allow acceptable marginal fit when complete seating is achieved. The shoulder-bevel margin is the least esthetic choice. The bevel should be used only with metal-ceramic crowns and is suited for structurally compromised teeth where ferrule extension is important. If porcelain is placed on a bevel, the cementation process may cause porcelain breakage. In two studies, the geometry exhibiting the least marginal discrepancy after cementation was a shoulder preparation, which was significantly better than that of a shoulder bevel or chamfer.[143,144] The shoulder finish line is thought to be better than the beveled shoulder because it allows the excess cement to escape more readily. The shoulder design exhibits less marginal distortion than a chamfer because of the crown's thickness adjacent to the margin. Stress analysis of various margin finish lines showed that chamfer and internally rounded shoulder preparations had the lowest stress concentration when loaded vertically, reducing the risk of catastrophic ceramic failure under load.[145,146] When utilizing metal-ceramic crowns, porcelain margins must provide an esthetic transition from tooth to crown, preventing the margin from becoming the visual focal point. Such margins are easier to fabricate and more predictable when they are fabricated on a 90-degree shoulder preparation.

This is true for metal-ceramic and most all-ceramic systems.[147–150] A chamfer requires minimal axial reduction and is appropriate for conservative all-ceramic restorations. It does not, however, provide an adequate reduction for metal-ceramic crowns. The opaque material used to mask the metal at the margin will show through the more translucent porcelain and will compromise the esthetics in the gingival third. A 90-degree shoulder between 1.0 and 1.5 mm in depth allows for a precise margin, maximum seating, and good esthetics.

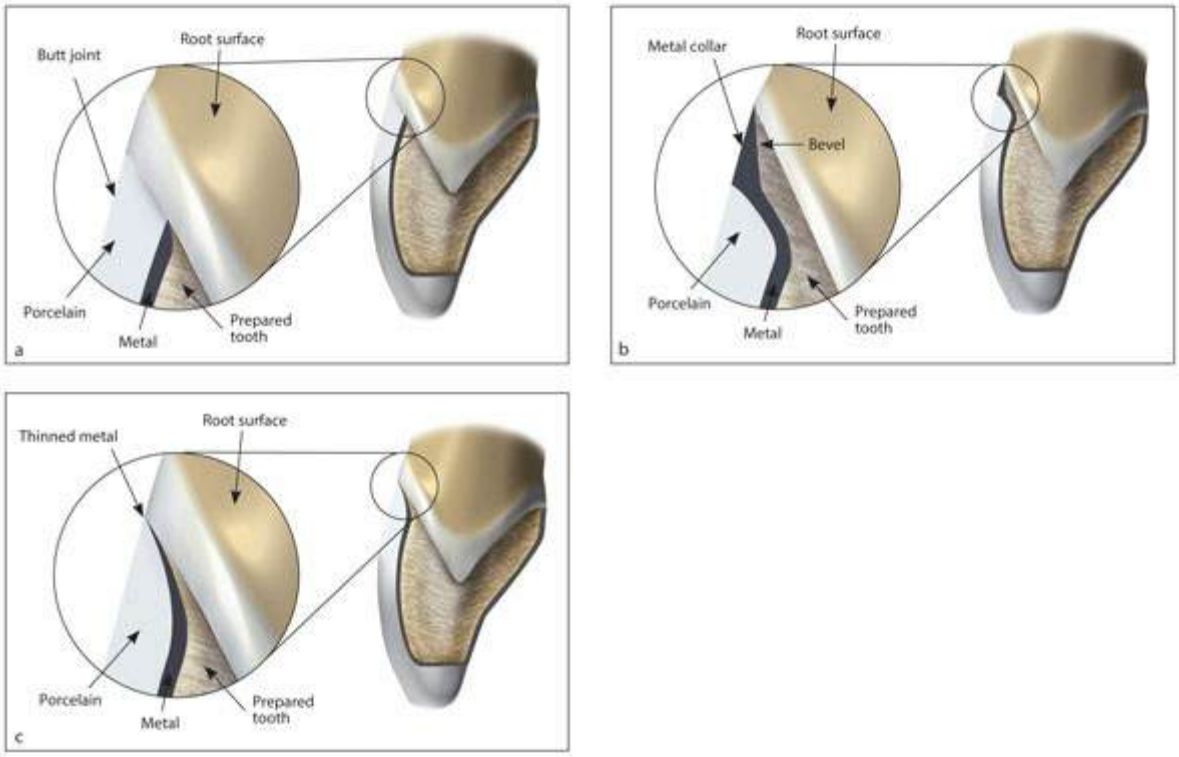



**Fig 18-25***(a)* In the preferred substructure design for a porcelain facial margin, a uniform thickness of porcelain is carried to the finish line. *(b)* The preferred substructure design if a metal facial margin is desired. Metal covers the bevel and forms a butt joint with the porcelain. *(c)* A 1.0-mm chamfer can be used for anterior all-ceramic restorations and metal-ceramic restorations. The gingival esthetics may be compromised with metal-ceramic restorations because of the extension of the metal to the margin.

The ideal finish line form and depth for all-ceramic systems vary between type of ceramic and fabrication technique. They all have two common elements: adequate axial depth and uniform smoothness.[151] A study on the effect of preparation design on the marginal integrity of Procera crowns highlighted this fact. It demonstrated that a feather-edge finish line produced crowns with over twice the marginal opening of a chamfer or rounded shoulder finish line. Furthermore, the deeper axially the rounded shoulder, the tighter the marginal fit.[54] Crowns created from IPS Empress CAD material exhibited similar results.[42] The best finish line is a rounded axial-

gingival line angle with a butt-joint external cavosurface margin. Ideal cervical preparation taper is imperative with all-ceramic crowns. Ceramic systems cannot accurately reproduce the contours of auxiliary grooves or deep occlusal morphology, eliminating their use as resistance features.[54] In addition, adequate occlusal/incisal reduction is more critical with all-ceramic than with metal-ceramic restorations. Inadequate space prevents the use of monoblock all-ceramic crowns and can compromise esthetics and strength.[152]

## Functional Crown Preparation Technique

With a complete understanding of the key components of intracrevicular tooth preparation, practitioners can achieve predictability with a standardized, controlled, and functional crown preparation.[153] This preparation design works for any type of anterior ceramic crown with modifications only in depth of preparation based on ceramic choice. This technique also minimizes the number of burs, bur changes, and cost. Only four burs are used in the technique sequence described below.

### Bone sounding

Before beginning any anterior crown preparation, a comprehensive evaluation of the underlying osseous structure is critical. Bone sounding to the osseous crest with a periodontal probe, as described previously in this chapter, should be performed to determine the dimension of the total dentinogingival complex. If the probing depth is 3.0 mm facially and 3.0 to 4.5 mm interproximally, the crest relationship is normal and tissue levels should remain stable. If, however, the probings are not normal, the patient must be made aware of the possible complications involved in the treatment, and the preparation and impression must be altered accordingly (Fig 18-26).



**Fig 18-26***(a)* Biologic dimensions around each anterior tooth must be evaluated. Visually, the tissue surrounding the preparations appears to be normal, but probing of the teeth demonstrates normal, high, and low osseous crest positions. *(b)* A 2-mm probing depth to bone facial to the left central incisor. Teeth with high crests are prone to biologic width violations. *(c)* Bone sounding on the left canine demonstrates a low crest. The tissue will be susceptible to recession.

## Reduction guides

When the teeth are severely worn or malpositioned or the vertical dimension is to be altered in the definitive restorations, the dentist will require a guide to allow proper reduction (Figs 18-27a and 18-27b). Multiple reduction-matrix designs have been proposed in the literature. Most require the tooth to be prepared, after which the matrix is placed to confirm that the tooth was reduced adequately. The problem is that there is no recovery when the practitioner removes too much tooth structure. A more accurate guide would encourage the proper depth of reduction in all dimensions without the risk of blindly overreducing the tooth. Aesthetic preevaluative temporaries (APTs)[154] were developed as a preoperative opportunity for the patient to evaluate the esthetics of veneer restorations. This same technique can be used as a guide for proper tooth reduction for an anterior crown. The APT is fabricated from a properly contoured diagnostic wax-up (Fig 18-27c). Condensation silicone is adapted directly on the cast. It is placed in a pressure pot at 20 psi for 15 minutes. This will compress the silicone into the contours of the wax-up, producing a highly accurate matrix. A notch is made between the central incisors to be used as a seating guide intraorally. Deep cuts on the buccal aspect extending to the teeth or vent holes through to the teeth are made in the silicone, creating a spillway for the excess acrylic resin (Fig 18-27d). Without this escape route, the acrylic resin will not flow, preventing the matrix from fully seating. If the matrix does not seat properly, it creates an error in the reduction. After the patient is anesthetized, the matrix is loaded with a provisional material (made from a bis-acrylic resin) and seated. After the material is allowed to set, the matrix is removed. The acrylic resin will remain on the teeth, engaging the embrasures (Fig 18-27e). There will be a variety of thicknesses of the acrylic resin depending on the amount of ideal tooth morphology missing. This is the proposed position and contour of the definitive restorations. Proper reduction cuts are made into the matrix (Fig 18-27f), and then the remaining resin is removed. Any bur cuts on the teeth indicate the location and depth of reduction that is necessary. If there is no mark on the tooth, it is already reduced enough for the definitive restoration (Fig 18-27g), although some preparation may be necessary to provide gingival margins and an appropriate path of draw.



**Fig 18-27***(a)* Significantly worn dentition requiring restoration at an increased vertical dimension of occlusion. *(b)* New vertical dimension is demonstrated on the articulated casts. *(c)* Missing tooth morphology is added in wax on the cast. *(d)* Silicone matrix for reduction guide and provisionals. The midline is marked, and vent holes are created on facial surfaces. *(e)* Bis-acryl resin is added to the matrix and seated completely. Once polymerized, the acrylic will remain on the teeth. This demonstrates the desired final contours and vertical alteration. *(f)* Incisal, facial, and lingual reduction cuts. *(g)* After removal of the remaining bis-acryl resin, tooth reduction can be visualized. Note that many teeth require little to no tooth removal for definitive restorations other than that required to establish gingival margins and a path of draw.

## Incisal reduction

The amount of incisal edge reduction is dictated by the planned final incisal edge position. There should be a 2.0-mm reduction from that point. This requires that a diagnostic wax-up be completed before tooth preparation if the original incisal edge position is not acceptable. A laboratory-based retrospective analysis of dies prepared in general dental practices for all-ceramic crowns concluded that 86% had been underreduced incisally.[155] A 2.0-mm reduction allows the technician space to develop incisal translucency and natural tooth characteristics without a loss of fracture strength of the porcelain.[156] Initial depth cuts are made with a no. 330 carbide bur that has a 2.0- mm cutting-head length (Fig 18-28); note that no. 330 burs from some manufacturers have a head length of less than 2.0 mm, so the head length should be measured. This same carbide bur can be used to open proximal contacts and is also excellent for use in occlusal reduction of posterior teeth. A 1.2-mm-

diameter round-ended cylinder diamond is used to remove the remaining incisal edge tooth structure to the level of the incisal depth cuts (Fig 18-29). This diamond is also used for facial and cervical reduction.



**Fig 18-28** Incisal reduction grooves are made and interproximal contact is broken with a no. 330 carbide bur with a 2.0-mm cutting-head length.



**Fig 18-29** A 1.2-mm round-ended cylinder diamond completes the incisal reduction.

## Facial reduction

Viewed from the proximal aspect, anterior teeth have three facial planes: cervical, midfacial, and incisal. Practitioners prepare teeth with little regard to this fact, and dies routinely demonstrate only one plane of reduction.[155] The incisal plane has been removed in the first step with a 1.2-mm-diameter round-ended cylinder diamond. The facial reduction focuses only on the midfacial plane, not the cervical. Depth cuts are prepared to the full depth of the diamond (Fig 18-30), which is aligned with the midfacial plane (Fig 18-31). The use of the 1.2-mm diamond as a depth gauge allows a more uniformly accurate preparation. Without the use of a device to help gauge reduction depth, teeth are routinely underprepared on the facial aspect.[157] At this time, the cervical plane should receive almost no reduction. If the bur is angled to the cervical aspect, the facial plane would be severely underreduced. After two or three depth cuts, the same diamond is used for the remaining gross facial reduction.



**Fig 18-30** Multiple facial reduction grooves are made to the entire depth of the 1.2-mm-diameter diamond bur.



**Fig 18-31** The facial reduction grooves must be oriented properly to ensure adequate midfacial reduction. The gingival plane is not prepared at this point.

## Cervical reduction

The cervical finish line is initially produced at the free gingival margin circumferentially (Fig 18-32). The tissue is not retracted prior to initial preparation. The primary cord can push the free gingival margin apically and flatten out the normal architecture up to 1.5 mm, which potentially increases the chances of biologic width violations at the line angles or interproximal areas (see Fig 18-18b). The 1.2-mm round-ended diamond is then used to create a finish line with a rounded internal line angle and slight lip rather than a flat shoulder. The recommended thickness is 1.0 to 1.5 mm depending on the type of crown chosen. Underreduction adjacent to the finish line will affect the emergence profile and facial contour of the restoration.



**Fig 18-32** The gingival third is reduced axially by 1.2 mm at the free gingival margin. At this point, the shoulder should be rounded internally, and a slight lip will be present at the outer edge of the finish line.

This can predispose the restoration to technical and/or esthetic failure. Utilizing a 1.2-mm-diameter diamond allows the practitioner a guide for preparation depth adjacent to the finish line. This is critically important because studies indicate a large (0.75 to 0.90 mm) variation between recommended tooth reduction and that actually achieved.[109]

## Lingual reduction

The finish line on the lingual is established at the free gingival margin with the same 1.2-mm-diameter diamond. Depending on the crown chosen, this finish line can be a light chamfer for metal, using up to half of the diameter of the diamond, or a 1.2-mm shoulder for cemented all-ceramic systems if the bur is used to its full depth. The proper reduction on the lingual concavity is required to ensure functional anterior guidance, prevent interference with the envelope of function, and avoid development of contours that produce unacceptable phonetics. Without proper reduction, technicians are forced to over-contour the lingual aspect of the definitive restorations. This can move the condyle distally and produce joint discomfort. In addition, a lisp after definitive restoration placement indicates an overcontoured lingual surface. Air flow becomes constricted, producing a whistle. A football-shaped diamond is designed to mimic the lingual concavity and provides the necessary reduction of 0.5 to 1.5 mm, depending on the choice of material used (Fig 18-33). With the initial cervical reduction completed, a primary thin retraction cord is placed in the sulcus.

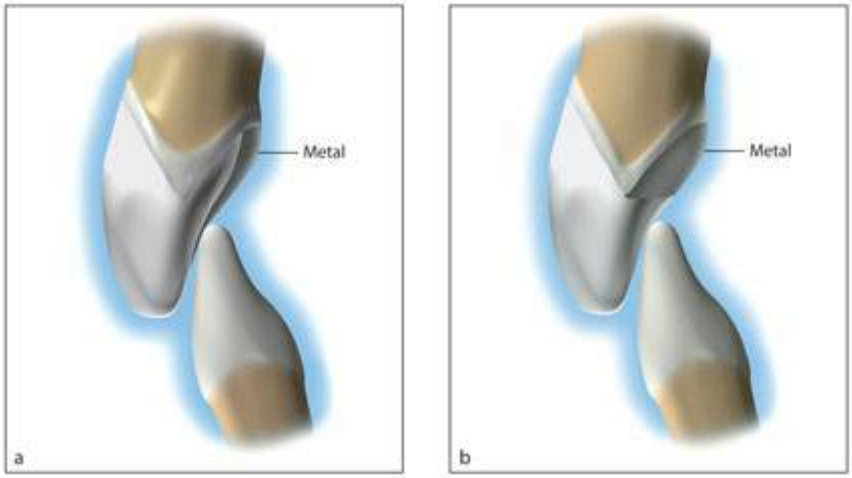



**Fig 18-33***(a)* In the preferred occlusal relationship between a maxillary metal-ceramic crown and a natural mandibular incisor, the contact is on metal. *(b)* In a less desirable occlusal relationship, the opposing natural incisor contacts the restoration on porcelain. This is the recommended occlusal relationship when both maxillary and mandibular incisors are restored.

## Preparation completion

The completion of the preparation begins by inserting a dry primary cord. The retraction of the tissue will allow unobstructed vision during margin refinement. A 1.6-mm-diameter round-ended diamond bur with 3 degrees of taper is used to finish the preparation. The taper on the diamond is designed to create a minimum convergence angle. The large tip will remove the J-lip on the external surface produced by the 1.2-mm diamond and create a crisp shoulder or chamfer circumferentially. A round-ended diamond is chosen because of its versatility and ease of use. The rounded tip can simply cleave the J-lip and produce a shoulder. If the side of the bur is used, it will remove the lip at an angle, producing a deep chamfer. The round end is suggested over a flat-ended diamond for fabricating a shoulder finish line. With a round-ended diamond, the practitioner can lock his or her wrist and allow the bur to simply traverse along the J-lip, thus producing the shoulder finish line. A flat-ended diamond requires a much more deft touch because the angle of the end must be altered constantly to match the contours of the preparation. Most practitioners find that the flat-ended bur produces an irregular, inadequate finish line. The margin should be developed to the coronal aspect of the primary cord facially and level with the free gingival margin interproximally (Fig 18-34). This will ensure an intracrevicular margin location after normal gingival tissue rebound, which takes up to 3 months in patients with normal crests. In patients with low or high crests, margins should not extend apical to the gingival crest.



**Fig 18-34** The marginal lip is removed with a 1.6-mm round-ended diamond bur, leaving a 90-degree shoulder.

## Impression Technique

An important factor in delivering high-quality restorations is providing the laboratory with an accurate impression. Practitioners must develop a technique that is reliable and repeatable.

Impression materials minimally displace gingival tissue, fluids, or debris, so successful retraction and isolation are mandatory. Various methods and techniques have been described in the literature for exposure of dry crown margins. They include mechanical, mechanical-chemical, rotary curettage, electrosurgery, and laser surgery (troughing). Regardless of the technique, four requirements must be satisfied for effective gingival retraction: *(1)* The tissue at the crown margins must be displaced horizontally to provide for an adequate bulk of impression material; *(2)* the tissue at the margin must be displaced vertically to expose the undercut apical to the finish line; *(3)* all gingival bleeding must be arrested; and *(4)* all hard and soft tissues must be dry.[158,159]

The mechanical-chemical technique, the most popular retraction technique today,[160,161] combines the mechanical displacement of the tissue with cord and the drying action of chemical agents. The retraction begins during crown margin preparation. The primary piece of dry cord (compression cord) is placed in the sulcus and cut so that the ends abut within the sulcus. After the final finish line preparation, aluminum sulfate gel is ejected around the sulcus. Aluminum sulfate is one of the least irritating of the retraction chemicals. It does not ionize readily, it retains its astringent qualities, and it is not toxic.[162] A larger, dry secondary cord (deflection cord) is packed through the aluminum sulfate, dragging the chemical into the sulcus. In studies by Laufer et al,[163,164] 50% to 90% of impressions of preparations with sulcular widths between 0.08 and 0.13 mm had defects. No

detectable difference in distortion of impressions of preparations was found with sulcular widths of 0.2 mm or greater. This was confirmed by Aimjirakul et al[165] in a gingival sulcus simulation model that demonstrated that the viscoelastic behavior of elastomers depends on the width of the sulcus. The polyether materials showed greater sulcular penetration than other impression materials, regardless of sulcular width, because of their greater hydrophilicity. To achieve a crevicular width of 0.2 mm, the secondary cord optimally must remain in the gingival crevice for 4 minutes prior to making the impression.[166]

Another difficult area for practitioners is managing the field. If the patient is involved in retracting the cheeks and lips and eliminating the parotid fluid flow, it frees the doctor and assistant to manage the tongue, saliva, sulcus, and impression. Absorbent parotid shields and cheek retractors are used to retract the lips and cheeks. The patient holds the retractors, leaving the operator's hands free (Fig 18-35). The double-packing technique, in a patient with a normal crest, should produce no apical migration of the gingival crest.[167–169] The technique must be modified for low- and high-crest patients. Finish lines should be kept supragingival or equigingival to avoid recession and biologic width impingements. The impression technique in a low-crest situation must use light packing pressure and smaller diameter cords to avoid trauma-induced recession. Rotary curettage, electrosurgery, and laser troughing are definitely contraindicated in low-crest patients. A high-crest patient simply lacks the depth for the normal retraction cords, and smaller cords are recommended. The secondary cord should be removed wet to prevent renewed hemorrhage. The preparation is washed and dried to evaluate tissue displacement and hemorrhage control.

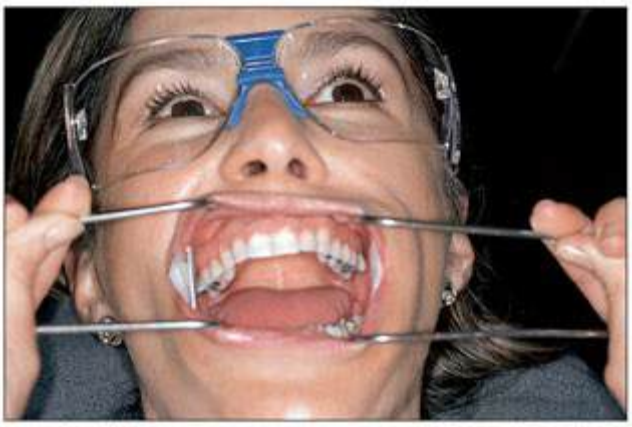


**Fig 18-35** The patient helps in the impression process by holding the cheek retractors, keeping the dentist's and assistant's hands free to work.

If the field is not dry, a 12.7% iron solution can be used to cauterize the tissue.

Ferric solutions provide minimal tissue injury, and healing is more rapid than with aluminum chloride.[162] It must be placed directly against the cut tissues because it coagulates blood so quickly; if not, it is washed away by the extraneous blood, leaving a bleeding site. There have been reports of the tooth absorbing the ferric ion. In vitro tests, however, have failed to show corrosive effects or staining.[162] A disadvantage to the use of ferric compounds is that they inhibit the setting of polyether impression materials. If a dry field cannot be achieved rapidly, the sulcus width will decrease, increasing the risk of defects in the impression. Chemical agents and secondary cord should be replaced and the retraction attempted again. Only after the field is acceptable is the impression material mixed or digital impression begun.

All impression materials are accurate enough to lead to production of well-fitting restorations.[170–172] Polyether and polyvinyl siloxane impression materials work well for multipurpose use and provide good soft and hard tissue detail.[173,174] Polyvinyl siloxane materials provide good elastic recovery; are dimensionally stable for indefinite periods; and are clean, odorless, and tasteless.[175] A disadvantage of polyvinyl siloxane is that because it is hydrophobic, it requires an absolutely dry field. Although many polyvinyl siloxane impression materials are advertised to be hydrophilic, research indicates that they are only reliable under dry conditions. In all situations, moisture leads to less detail reproduction compared with dry conditions.[176]

Latex gloves may inhibit the set of polyvinyl siloxane. Even casual contact of the teeth or soft tissues with latex gloves can cause an inhibition of polymerization intraorally.[177–179] (Fig 18-36). Latex gloves do not affect polyether impressions. Polyether materials are inherently hydrophilic, so small amounts of blood or saliva do not affect accuracy.[176] Polyether is affected by ferric solutions and some disinfecting solutions. Because of its rigid set, it can be difficult to remove from the mouth, and it has a bad taste and smell.[170,171] Polyvinyl siloxane and polyether materials can be used in a custom or stock tray, and studies have shown no difference in accuracy.[180,181] The working and setting times of the impression materials must be identified and followed precisely.



**Fig 18-36** The staining of a latex glove indicates inhibition of polymerization of polyvinyl siloxane impression material.

For years, casts made from traditional impressions have been digitally scanned and copings fabricated from a virtual file. Today, these digital impressions can also be made intraorally, reducing the additional error associated with cast fabrication.[182] Digital impressions address many of the material-handling difficulties of traditional impressions (Fig 18-37). They do not, however, relieve the practitioner of the necessity of proper isolation and management of the field.[182] In fact, it makes these tasks even more critical. Many dental manufacturers have introduced digital impression systems (see chapter 19). Some scanning systems are part of a ceramic milling system designed to produce a crown or veneer virtually. A variety of ceramic and resin choices can be produced in the office or "shipped" virtually to a laboratory for fabrication.[183] Other technologies simply create a resin cast from which any laboratory procedure can be accomplished. While the research is currently rather limited on the accuracy of the images, it appears that digital replicas are well within the tolerances required of impression materials. An in vitro analysis was conducted on the cross-arch accuracy of digital impressions using conventional impressions as the control. Two different intraoral digital scanners demonstrated trueness and precision equivalent to that of the conventional techniques.[184] Currently, no one system appears to satisfy the needs of every practitioner or produce an accurate implant platform transfer. It does appear that the digital impression will begin to minimize the use of traditional impression materials in the near future.

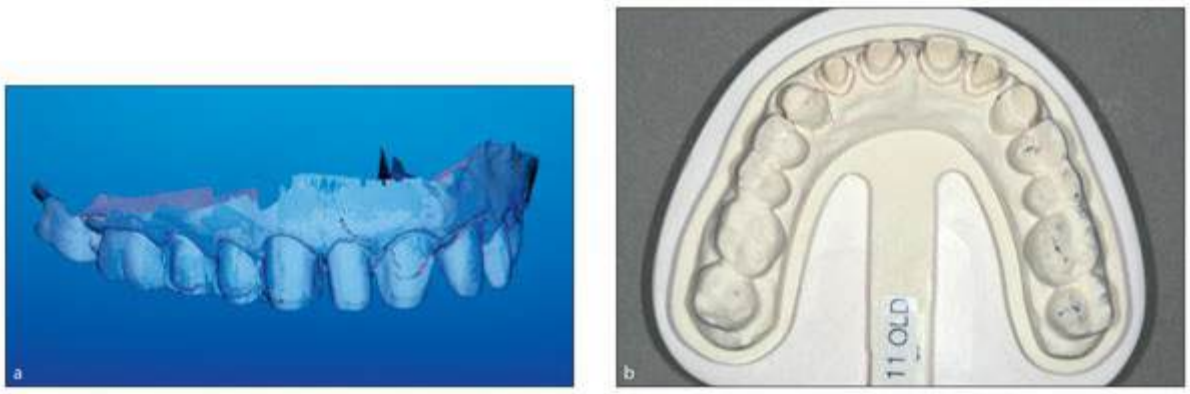



**Fig 18-37***(a)* The intraoral digital image of prepared teeth. *(b)* The cast produced from digitized data.

## Shade

Success of an anterior ceramic crown depends on how well critical information is communicated between the dentist and the laboratory. At least 75% of all remakes are caused by poor communication.[152] It is the dentist's challenge to predictably and accurately convey what is visualized in clear terms, allowing the technician to develop the "illusion of reality" in the ceramic material. The dentist must be able to perceive the color, understand and quantify the characteristics, and convey the information.

The experience of seeing color in teeth depends on several factors,[185] as defined in chapter 4.

The shade guide is less important than the understanding that the underlying color of the preparation may have a dramatic impact on the final esthetic results. Ceramists can adjust the ceramic choice, core color, translucency, and veneering ceramic to modify the crown to "block out" discoloration. This cannot be accomplished without communicating the preparation color information to the laboratory. Unfortunately, 93% of clinicians in a recent review of general practices failed to provide any information regarding the shade of the prepared tooth[155] (Fig 18-38).




**Fig 18-38** Use of a stump shade guide is demonstrated in a photograph for the ceramist.

Communication between the dentist and the laboratory can occur in four ways:

1. *Diagrams:* A representation of the tooth may be drawn. While this can be helpful for communicating difficult contours and characteristics, it is difficult for the technician to interpret and very time-consuming.
2. *Custom stain tabs:* Painting colors on a shade tab can display chroma but cannot represent value. In addition, customizing is extremely time-consuming.
3. *Photographs:* Taking photographs of the teeth being restored along with matching shade guides is the most efficient and cost-effective method of shade selection (Fig 18-39).




**Fig 18-39** Photographs for shade matching should be made at an angle to avoid excess glare on the tooth surface. The shade tab should be held in line with the long axis of the tooth.

4. *Colorimeter and spectrophotometer:* A digital analysis of color may be helpful but is still not reliable enough to create predictable results in clinical practice.

Photographs must provide an accurate representation of nature. A 35-mm digital

camera with neutral slide film (EPN 100, Kodak) maximizes the ceramist's ability to mimic reality. Digital photography has begun to replace film cameras. The ability to instantly view results and the elimination of film processing are tremendous advantages. Electronic transfer to the laboratory allows improved communication. The disadvantage is the multiple errors that can occur in color reproduction. The camera and monitors upon which the images are viewed must all be color-corrected to maximize results. Many operatory and laboratory shade-matching systems have been introduced with varying success.

The more photographs that are taken, the more information the technician can obtain. If the anterior ceramic crown does not match at try-in, more photographs should be taken of the crown to allow the ceramist to modify it correctly. Personal contact between the ceramist and the patient would be ideal, but this often is not possible.

Despite repeated experiments detailing the inaccuracy of the traditional visual shade-matching technique, digital analysis of color has not made significant inroads into the profession. Colorimeters and spectrophotometers have the potential to improve shade matching by eliminating the significant subjective factors and examiner variability. Research has emphasized two problems with color selection devices: device accuracy and tooth morphology. Most shade-matching devices have a high degree of reliability, but there is significant variability in the accuracy of the commercially available machines. Shade-matching systems have been reliable but variable between devices at matching shade guides. This means that the system can produce the same information on repeated measurements; however, it does not address the accuracy of the information.[186] Measuring the shade of homogenous porcelain shade tabs is certainly simpler than measuring the shade of a complex human tooth. The polychromatic color, surface texture, and varying translucency make intraoral applications more circumspect. Differences in tooth contour will also make positioning difficult. Researchers studied the intradevice repeatability and interdevice reliability between two shade-selecting devices in measuring the shade of natural dentition. The results showed a significant intradevice difference for all color parameters. The interdevice agreement rate for the systems on matching shade tabs was also moderate. The complexities of the human dentition apparently still pose a problem that digital recordings have yet to overcome.[187]

## Restoration Placement

The classification of anterior ceramic crowns is based on whether the prosthesis is

adhesively retained or cemented. Veneers and bonded crowns rely on the formation of a resin bond between the tooth structure and porcelain. Cemented all-ceramic and metal-ceramic crowns are typically nonadhesively cemented with traditional cements rather than bonded to the preparation. Therefore, the placement protocol will vary significantly between the types of crowns.

## Adhesive bonding of all-ceramic crowns

The goal of adhesive bonding is to provide a marginal seal of the crown and adhesively retain it to the tooth. Adhesive bonding is technique sensitive and demands proper preparation of ceramic and tooth surfaces. It has been demonstrated that a strong, dependable bond between resin and porcelain can be achieved. The porcelain intaglio surface is etched with hydrofluoric acid to create micromechanical retention sites.[188] Silane is added to the etched surface shortly before bonding and allowed to air dry. Silane coupling agents create a chemical bond of the adhesive resin to porcelain.[189]

The dentin-resin bond is less dependable than the resin-ceramic bond. Because hemorrhage and crevicular fluid flow may interfere with dentin bonding, teeth should be isolated with retraction cord before cementation. Preparations should be cleaned with pumice and/or antimicrobial solutions, such as chlorhexidine. Because light polymerization decreases with increasing porcelain thickness[190] and because polymerization of the adhesive before cementation may result in resin pooling and incomplete seating, resin cements must reach final polymerization with reduced or even no benefit of light activation, depending on the opacity and thickness of the ceramic restoration. This necessitates the use of dual-curing resin luting agents to ensure that the cement will polymerize with limited light exposure. However, dual-curing resin materials should always be light cured, because any light exposure will enhance the degree of conversion of a dual-curing material.[191]

Manufacturers have provided three alternatives for dentists for developing adhesion between resin cements and tooth structure: etch-and-rinse and self-etching adhesives and self-adhesive cements.[192] The *etch-and-rinse system* includes an etchant, primer, and dual-curing adhesive. A dual-curing resin cement is recommended for bonding of the crown for the same reason. Dual-curing adhesives have a slow, chemically activated autocuring component and a light-activated component. The inside of the crown is painted with adhesive before addition of the luting resin. The crown is gently placed and excess cement removed with a brush. Then the crown is seated with additional pressure or lightly tapped to extrude excess

luting resin at the margins. The luting resin is light cured through the facial and lingual aspects for 1 minute each. Excess resin is removed with a no. 12 or 12B scalpel blade. The occlusion is adjusted under water spray with diamond burs that have 15-and 8-μm diamond abrasive. Finally, the porcelain is polished with a porcelain polishing kit. This technique provides the highest cement-to-tooth bond but requires the most steps and has the greatest risk of field contamination.[192] *Self-etching systems* provide a self-etching primer to prepare the tooth. The cement is mixed, added to the crown, and seated over the preparation. After the components are chemically cured, the excess cement is removed with a scaler. The etch-and-rinse systems provide a more reliable bond and possibly a more stable bond than the self-etching technique.[193] The *self-adhesive systems* have a weak acid etchant, primer, and adhesive incorporated into the cement, appreciably reducing the risk of contamination. The bond strengths vary notably between systems.[192] Because of the nature of the chemistry, the etchant is not appropriate for enamel preparations. Some self-adhesive resin cements demonstrate high strength at baseline but lose strength dramatically over the first few months when compared with self-etching resin cements.[194]

## Cementation of nonbonded crowns

The goals of nonadhesive cementation are complete seating of the crown and maximization of the physical properties of the cement. Classically, zinc phosphate was the cement of choice. Today, alternatives provide increased compressive strength and limited marginal dissolution. They include glass-ionomer and resin-modified glass-ionomer cements. The preparation should be thoroughly cleaned with pumice and/or an antimicrobial solution. If the soft tissue interferes with complete seating, retraction cord should be placed. The cement should be mixed according to the manufacturer's instructions and loaded into the crown, and the crown should be seated. If the crown has metal margins, the patient should bite on an orangewood stick or cotton roll; the stick can be moved up and down and back and forth for a few seconds. This technique, called *dynamic seating*, results in more complete seating of the crown.[195] Because of the risk of fracture, a cotton roll is recommended in place of the stick for all-ceramic crowns. Every effort should be made to keep the crown dry during the initial setting phase, that is, the first 5 minutes after cementation.[196] The physical and mechanical properties of most cements deteriorate if they become wet during setting. The final step is careful removal of the excess set cement with an explorer or scaler. This class of cement will not adhesively bond to most ceramic

systems.[197] Because of this lack of bond, the ceramic may fatigue quicker and be more prone to fracture.[198] However, conventional cements are recommended for high-strength ceramics, such as alumina or zirconia.

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